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ESSENTIALS OF IRON ORE DEPOSITS

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ESSENTIALS OF IRON ORE DEPOSITS

Dr. Hassan Z. Harraz

hharraz2006@yahoo.com

Autum 2023

“Iron Ore is more Integral to the Global Economy than any Other Commodity, Except Perhaps Oil”.

Christopher LaFemina, mining analyst at Barclays Capital

(In 2011 the Financial Times quoted)



OUTLINES

- This Introduction to Iron ore overview includes:

1. INTRODUCTION

- 1.1. The Global Value of Iron Ore**
- 1.2. Value of Local Industry**
- 1.3. Principal Iron-bearing Minerals**

2. IRON ORE DEPOSITS

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- 3.1. Hematite**
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 - iv) Flotation**
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 - vii) Lump, Fine, and slime iron ores**
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10. ABOUT IRON ORE MARKET

11. Conclusion

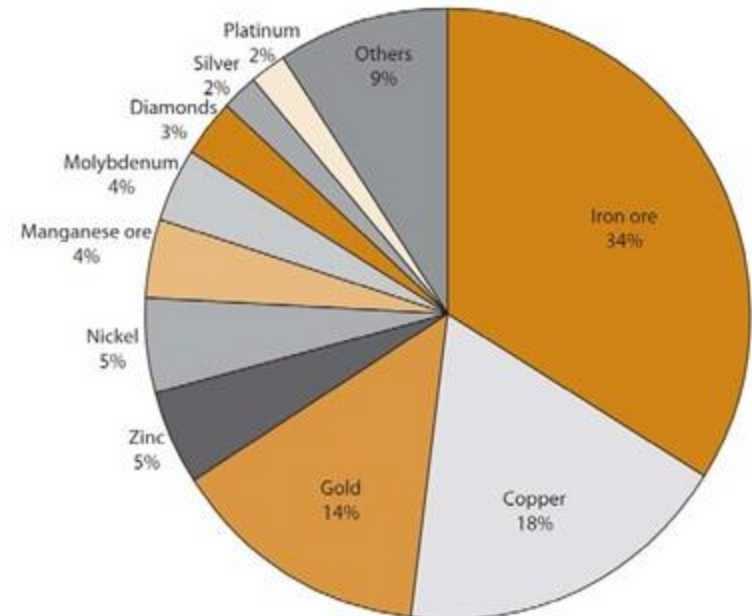
1. INTRODUCTION

1.1. The Global Value of Iron Ore

- ❑ Besides the fact that steel is a basic infrastructural input throughout every developing and developed economy in the world, the volume and value of iron and steel production is often overlooked. The value of iron ore represented 34% of the global mineral production in 2009 (not including the energy commodities, coal, oil and gas) as can be seen in Figure 1.

1.2. Value of Local Industry

- ❑ The value of a local steel industry to the economy cannot be over-emphasized.
- ❑ Whilst the production of steel is not always a highly profitable industry in itself, it provides many jobs, has a very high multiplier effect and opens the door to a vast array of associated upstream, sidestream and especially downstream linkages.



Source: Raw Materials Data as shown in ECA 2011.
Note: Inclusive of minerals only (energy commodities not included)

Figure 1: Value of global mined minerals 2009

1.3. Principal Iron-bearing Minerals

Iron (Fe) is a metallic element that constitutes 5.6% by weight of the Earth's crust and is the fourth most abundant element in the crust.

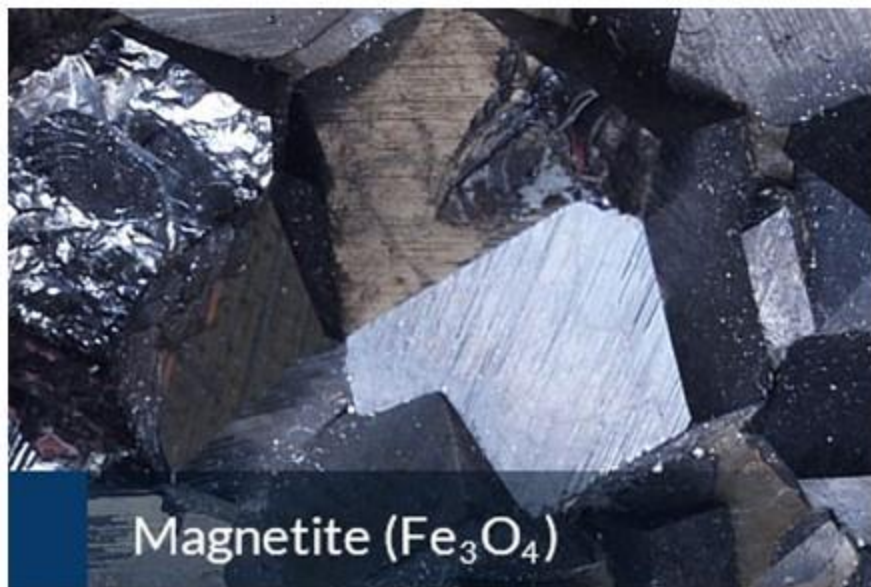
Over 300 minerals contain iron but five are the primary sources of iron-ore minerals: magnetite (Fe_3O_4), hematite (Fe_2O_3), goethite ($\text{Fe}_2\text{O}_3\text{H}_2\text{O}$), siderite (FeCO_3), pyrite (FeS_2). The first three are of major importance because of their occurrence in large economically minable deposits.

Table 7.2 Principal iron-bearing minerals in iron-rich sedimentary rocks

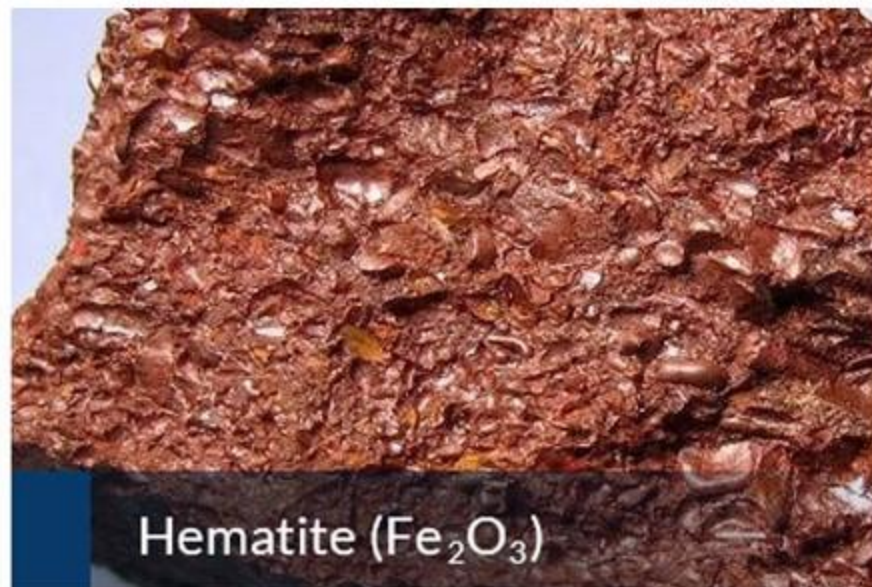
Mineral class	Mineral	Chemical formula
Oxides	Goethite*	FeOOH
	Hematite	Fe_2O_3
	Magnetite	Fe_3O_4
Silicates	Chamosite	$3(\text{Fe, Mg})\text{O} \cdot (\text{Al, Fe})_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot n\text{H}_2\text{O}$
	Greenalite	$\text{FeSiO}_3 \cdot n\text{H}_2\text{O}$
	Glaucconite	$\text{KMg}(\text{Fe, Al})(\text{SiO}_3)_6 \cdot 3\text{H}_2\text{O}$
	Stilpnomelane	$2(\text{Fe, Mg})\text{O} \cdot (\text{Fe, Al})_2\text{O}_3 \cdot 5\text{SiO}_2 \cdot 3\text{H}_2\text{O}$
	Minnesotaite (iron talc)	$(\text{OH})_2(\text{Fe, Mg})_3\text{Si}_4\text{O}_{10}$
Sulfides	Pyrite	FeS_2
	Marcasite	FeS_2
Carbonates	Siderite	FeCO_3
	Ankerite	$\text{Ca}(\text{Mg, Fe})(\text{CO}_3)_2$
	Dolomite	$\text{CaMg}(\text{CO}_3)_2$
	Calcite	CaCO_3

*Not found in Precambrian iron formations

From Boggs, Principles of Sedimentology and Stratigraphy, 4th ed., p. 219



Magnetite (Fe_3O_4)



Hematite (Fe_2O_3)



Limonite ($2\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$)



Siderite (FeCO_3)

2. IRON ORE DEPOSITS

- ❑ Iron ore mineral deposits are widely dispersed in the worlds and form in a wide variety of geologic environments, including sedimentary, metamorphic, and igneous rock formations.
- ❑ Iron ore deposits are formed by three geologic processes:

Geological Type	Principal Iron Mineral	Average Iron Content (%)
SEDIMENTARY DEPOSITS		
Banded Iron-Formations (BIFs)	Magnetite; Hematite; Siderite; Iron silicates	33
Ironstone	Limonite; Hematite; Siderite; Chamosite	30
IGNEOUS ACTIVITY		
Magmatic Segregations	Titaniferous; Magnetite; Ilmenite; Iron silicate	65
Pyrometasomatic	Magnetite	45
SURFACE AND NEAR SURFACE WEATHERING		
Secondary enrichments of low-grade iron deposits	Limonite; Hematite; Siderite	55

2. IRON ORE DEPOSITS (cont.)

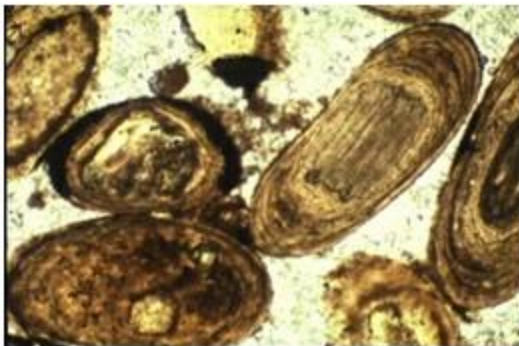
Economic iron ore deposits occur in two natures:

Ironstone (oolitic) Deposits

- are dominantly Phanerozoic sedimentary deposits that form poorly-banded or nonbanded bedded successions.
- commonly have an oolitic texture and may contain fossils that have been partially or completely replaced by iron.
- Sedimentary structures (i.e. cross-bedding) are common.
- All ore minerals in these deposits are oxides and hydroxides
- The most common iron ore minerals are hematite, lepidocrocite and goethite

Bog Iron ore

- was formerly mined in Europe hundreds of years ago when the large iron ore deposits had not been discovered.
- consists of iron hydroxide (goethite) deposited in swamps and lakes as a product of bacterial action.



3. IRON ORES

- ❑ Despite being the fourth most abundant element in the Earth's crust, iron is one of the most important elements ever discovered.
- ❑ Iron ore is a combination of minerals from which metallic iron can be extracted on an economic basis.
- ❑ Iron ores are rocks and minerals from which metallic iron can be economically extracted.
- ❑ The ores are usually rich in iron oxides and vary in colour from dark grey, bright yellow, or deep purple to rusty red.
- ❑ The five main types of iron ore forming minerals are:↓

<i>Name of Mineral</i>	<i>Chemical Formulae</i>	<i>Fe (wt%)</i>
Magnetite	Fe_3O_4	72.4
Hematite	Fe_2O_3	69.9
Limonite	$(\text{FeO}(\text{OH}) \cdot n(\text{H}_2\text{O}))$	55.0
Goethite	$\text{FeO}(\text{OH})$	62.9
Siderite	FeCO_3	48.2



List of Iron Ores

3. IRON ORES

- ❑ An ore's quality is also impacted by the commercially worthless material surrounding it, collectively known as gangue. **Silica and phosphorus**-yielding compounds are especially important because *they affect the structure of the metal and can sometimes complicate the steel-making process.*
- ❑ Ores containing very high quantities **of hematite** or **magnetite**, typically greater than about 60% iron, are regarded as **high-quality iron ore** or **Direct Shipping Ore**, and can be fed directly into Ironmaking Blast Furnaces (BFs).
- ❑ Iron ore is the raw material used to make **Pig Iron**, which is one of the main raw materials further used to make **Steel** (~98% of the mined iron ore is used to make steel) other uses of iron ore are like industrial finishes, polishing compounds and sponge iron industries.

3.1. Hematite:

- ❑ It is most important iron ore mineral main source for industries. Its composition is Iron Oxide and sometimes slight amount of titanium. Its name comes from the *Greek word for blood, haima*, because of its reddish colour. Crystals occurs in thin plates, as well as bundles of small micaceous plates, and in thin splinters.
- ❑ Most commonly massive, mammillary, botryoidal, reniform, oolitic, stalactitic, and radiating. Scalenohedral and rhombohedral crystals occur, although infrequently, and dendritic and rosette forms are also found.
- ❑ Hematite may also form as a pseudomorph of other minerals, especially as octahedral crystals of Magnetite.
- ❑ Striking features are reddish streak.
- ❑ Its specific gravity is 4.9 to 5.3 gm/cm³ and luster is metallic to dull.
- ❑ Hematite is weakly magnetic and Para-magnetism.
- ❑ The content of iron in the ore and physical characteristics vary from place to place in different types of ores.



3.2. Magnetite:

- ❑ Magnetite, also referred to as **lodestone**, is a black isometric mineral best known for its magnetic properties.
- ❑ After hematite, it is second most abundant Iron bearing ore.
- ❑ The main iron mineral in magnetite ore is the ferrous iron oxide magnetite (Fe_3O_4).
- ❑ Black magnetic oxide of iron crystallizing in the isometric (Cubic) system with hardness of 5.5 to 6.5.
- ❑ Magnetite is commonly black, brown, or dark grey in appearance, sometimes with a deep purple hue, opaque and metallic luster.
- ❑ Magnetite ore is of little value in its raw state, but it offers considerable advantages in its concentrated form. These include providing a viable ironmaking commodity for premium quality steel production. By comparison, magnetite ore typically has much lower iron content when mined of between 25% and 40% Fe and in this form is unsuitable for steelmaking.
- ❑ Magnetite ore requires complex processing to separate magnetite minerals from other minerals in the ore to produce an almost pure magnetite concentrate with an iron content of between 68% Fe and 70% Fe that is highly sought after by steel makers.
- ❑ It occurs as a replacement product in sedimentary or metamorphic rocks. It is also found as placer deposits as -black sand in beach deposits and as banded layers in metamorphic and igneous rocks.

3.3. Limonite :

- ❑ **Limonite** is a brown iron ore formed due to wearing of iron minerals. Limonite includes amorphous Fe-hydroxides with variable amounts of water.
- ❑ **Limonite** is found in many rocks that give the brown, yellow, or tan colour.
 - It typically develops as a product of chemical weathering of many minerals containing Fe^{2+} or ferrous iron. In the oxidation process it oxidizes in the trivalent iron Fe^{3+} , which is known as a process of limonitization.
- ❑ Limonitization is well visible in yellow-brown colour in freshly broken off greenish-gray rocks along and around cracks.
 - The brown, yellow, or reddish-brown colour of clays and many other rocks is derived from limonite.
 - Limonite above the hematite and siderite deposits often forms a crust of weathering and known as the “iron hat.” ***Limonite and goethite are important sources of low-grade iron and nickel ore.***
- ❑ **Ocher or ocher** is a natural clay earth pigment that is a mixture of ferric oxide and varying amounts of clay and sand. The color ranges from yellow to deep orange or brown.

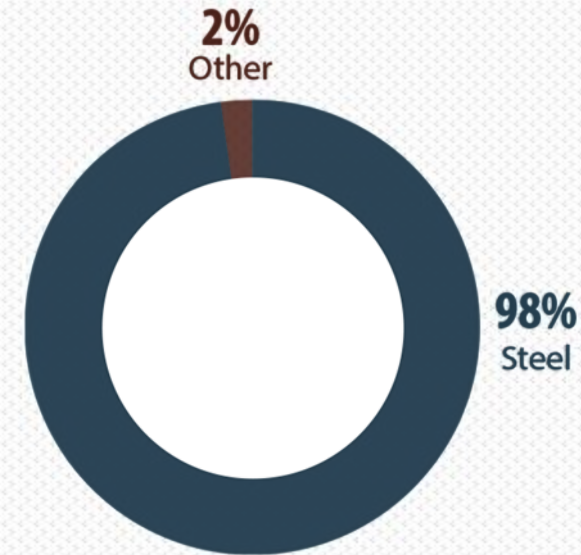
4. USES

❑ Iron ore is the base metal used to make steel – and steel is the most commonly used alloy in the world.

➤ From its earliest known uses as a metal in ancient times, iron ore has since undergone a transformation from a relic of the industrial age to the driving force behind a commodity super cycle that has reshaped the global economy.

❑ The primary use of iron ore (98%) is to make steel.

❑ The remaining 2% is used in a various other applications, such as powdered iron for certain types of steel, auto parts and catalysts; radioactive iron for medicine; and iron blue in paints, inks, cosmetics and plastics.



Iron ore, global uses, 2021

5. THE ESSENTIALS OF IRON ORE MINING

- ❑ Iron ore is a crucial natural resource that plays a fundamental role in the production of steel, a vital material for various industries such as construction, manufacturing, and transportation.
- ❑ Due to its widespread usage and economic importance, iron ore mining has become a significant activity worldwide.
- ❑ In this presentation, I will explore the essentials of iron ore mining, covering the process from extraction to processing.
- ❑ The mining of iron ore holds immense significance in both historical and modern contexts. Historically, iron ore has been mined for centuries, with evidence of iron smelting dating back to ancient civilizations such as *the Hittites in Anatolia and the Zhou Dynasty* in **China**.
- ❑ Iron ore mining has contributed to the development of numerous societies and has played a crucial role in the advancement of technology, infrastructure, and industrialization.
- ❑ In the present day, iron ore mining continues to be a vital economic activity. **Steel**, which is primarily produced from iron ore, is essential for constructing *buildings, bridges, and infrastructure projects*.
- ❑ Additionally, steel is a key component in the manufacturing of machinery, vehicles, appliances, and a wide range of consumer goods. The demand for iron ore remains high, driven by rapid urbanization, industrial growth, and the expansion of global trade.

5.1. Methods of Extraction

- ❑ The extraction of iron ore involves several stages, starting with the identification and location of iron ore deposits. Once the deposits are identified, various methods are employed to extract the ore for further processing.
- ❑ Iron ore can be extracted through different mining techniques, depending on the depth and accessibility of the ore deposit.
 - The two primary methods of extraction are open-pit mining and underground mining.

i) Open-Pit Mining

- ❖ Open-pit mining is the most common method used for extracting iron ore. This method involves the excavation of large open pits or quarries where the ore is relatively close to the surface.
- ❖ Heavy machinery, including excavators, trucks, and loaders, is used to remove the overlying layers of soil and rock to expose the ore. The extracted ore is then transported to the processing plant for further treatment.

ii) Underground Mining

- ❖ Underground mining is employed when the ore deposit is located at a considerable depth below the surface. This method involves the construction of tunnels and shafts to access the ore.
- ❖ Miners use specialized equipment and techniques to extract the ore, which is then transported to the surface for processing.





East Aswan Iron Ore, Egypt



5.2. Recovery and Transportation

- ❑ Once the iron ore is extracted from the mine, it goes through a process of recovery and transportation. The ore is crushed into smaller pieces to facilitate handling and transportation.
- ❑ It is then loaded onto **trucks, trains, or ships** and transported to the **processing plant** or **port** facilities for further processing and **export**.

5.3. Beneficiation and Processing of Iron Ore

- ❑ The processing of iron ore involves several stages to extract and refine pure iron.
- ❑ These stages include: *crushing and screening, grinding, magnetic separation, flotation, drying and sintering, and pelletizing.*

i) Crushing and Screening

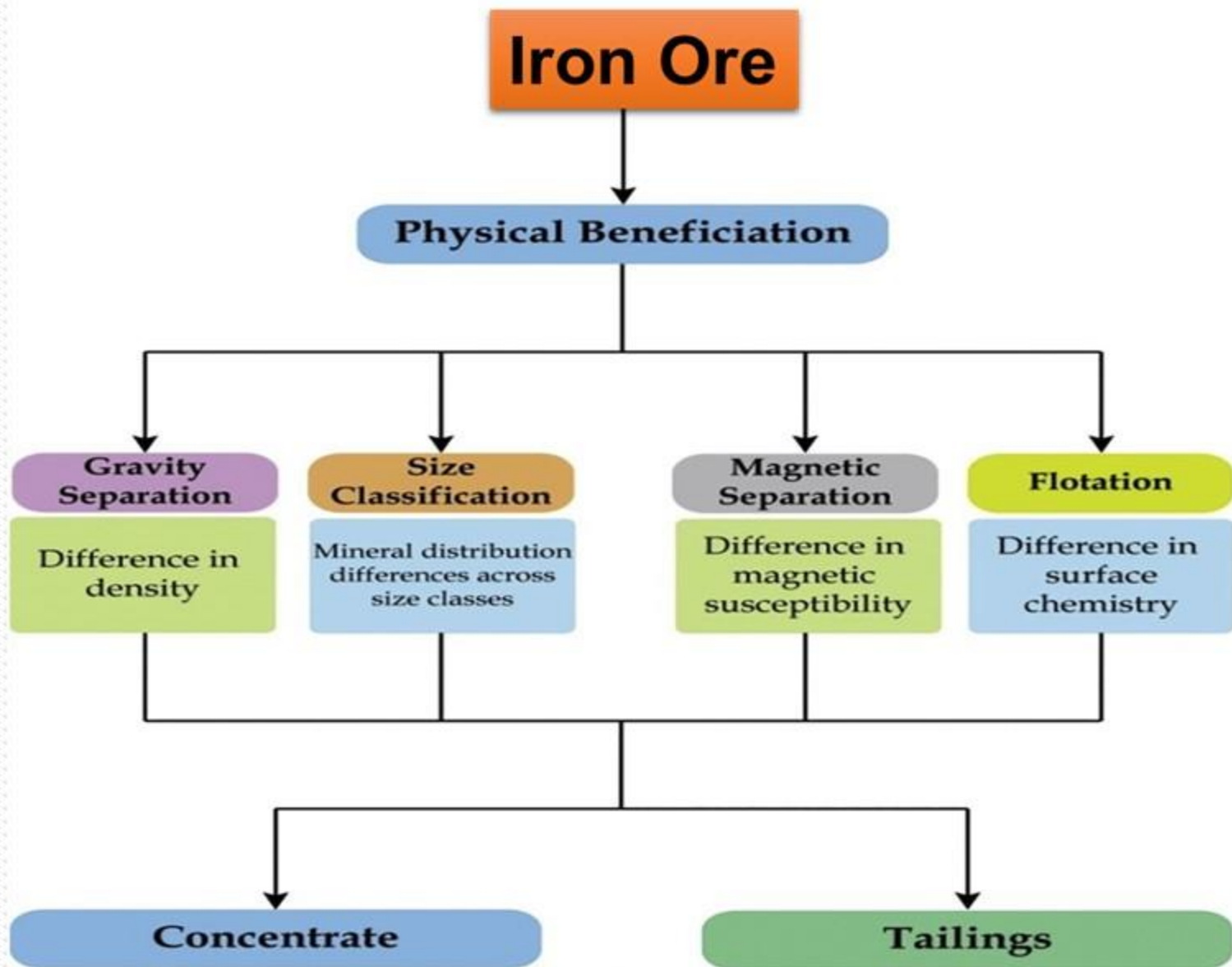
- ❖ The mined iron ore is crushed into smaller sizes and then screened to separate the different grades of ore. This helps in optimizing the efficiency of subsequent processing stages and ensures that only the desired size range is processed further.

ii) Grinding

- ❖ The crushed ore is further processed through grinding mills, where it is reduced to a fine powder.
- ❖ Grinding enhances the liberation of valuable iron minerals from the surrounding rock, allowing for better separation during subsequent processing.

iii) Magnetic Separation

- ❖ Magnetic separation is a commonly used method to separate magnetic minerals from non-magnetic ones.
- ❖ In iron ore processing, magnetic separators are used to separate and concentrate magnetic minerals (such as magnetite) from the non-magnetic components.



A schematic diagram of physical beneficiation process for the production of pellet feed concentrate from iron ores.

5.3. Beneficiation and Processing of Iron Ore...(Cont.)

iv) Flotation

- ❖ Flotation is a process used to separate minerals from the gangue (unwanted materials) based on their differences in surface properties.
- ❖ In iron ore processing, flotation is often used to separate silica and other impurities from the iron ore concentrate.

v) Drying and Sintering

- ❖ After the iron ore has undergone the aforementioned processes, it may be necessary to dry the concentrate for transportation or storage.
- **Drying** removes excess moisture from the concentrate, making it more manageable and reducing the risk of deterioration.
- **Sintering** is a process used to agglomerate fine particles into larger *lumps or pellets*. In iron ore processing, sintering involves heating the concentrate to a high temperature, causing the particles to bond together and form porous pieces that are more suitable for transportation and further processing.↓

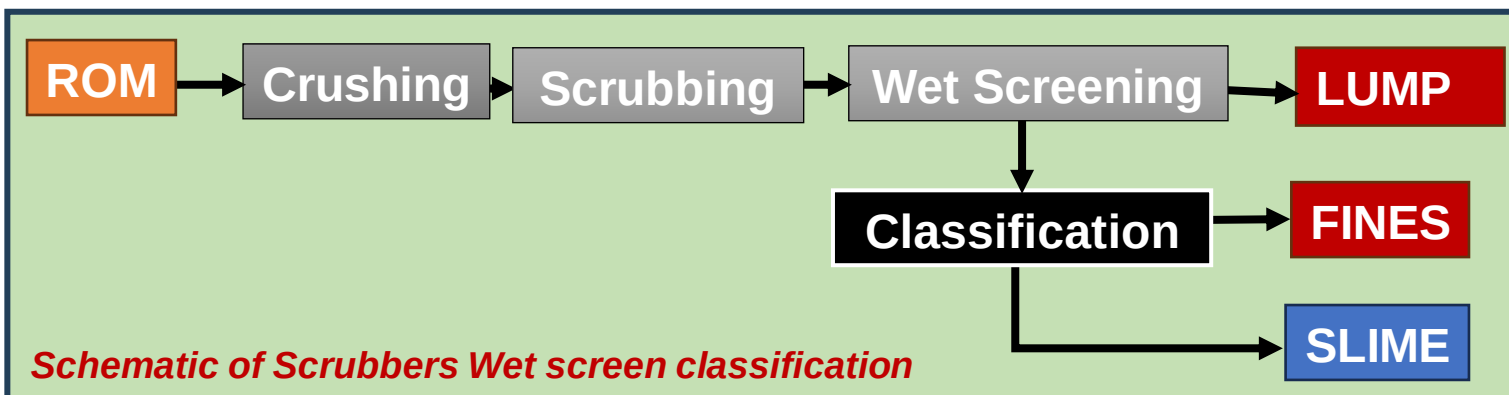
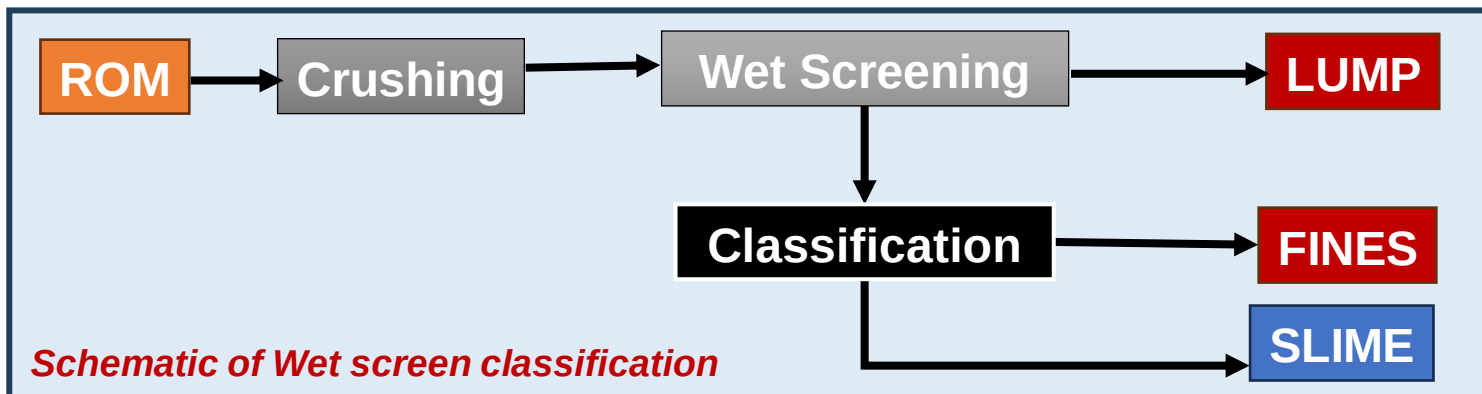
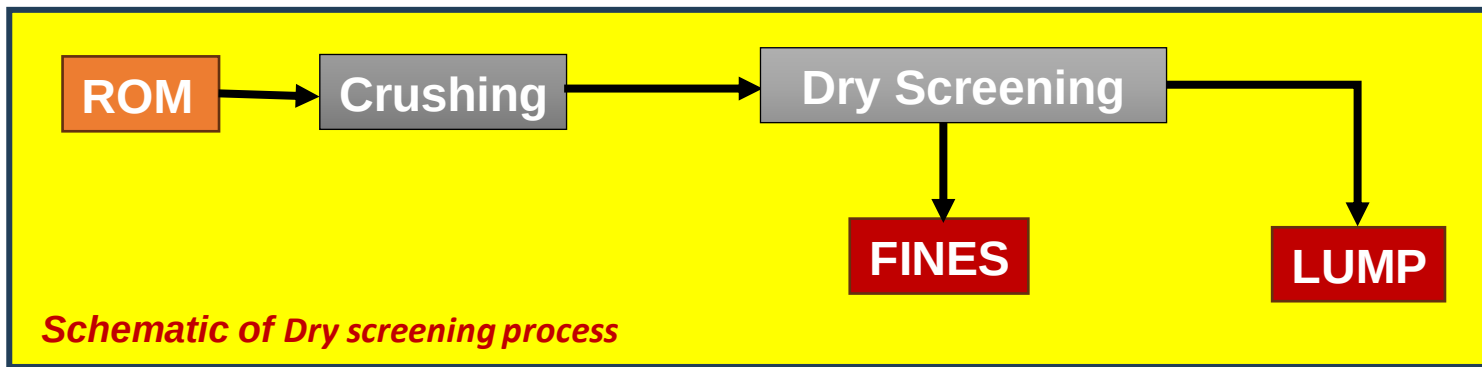
5.3. Beneficiation and Processing of Iron Ore ... (cont.)

Drying and Sintering, namely:

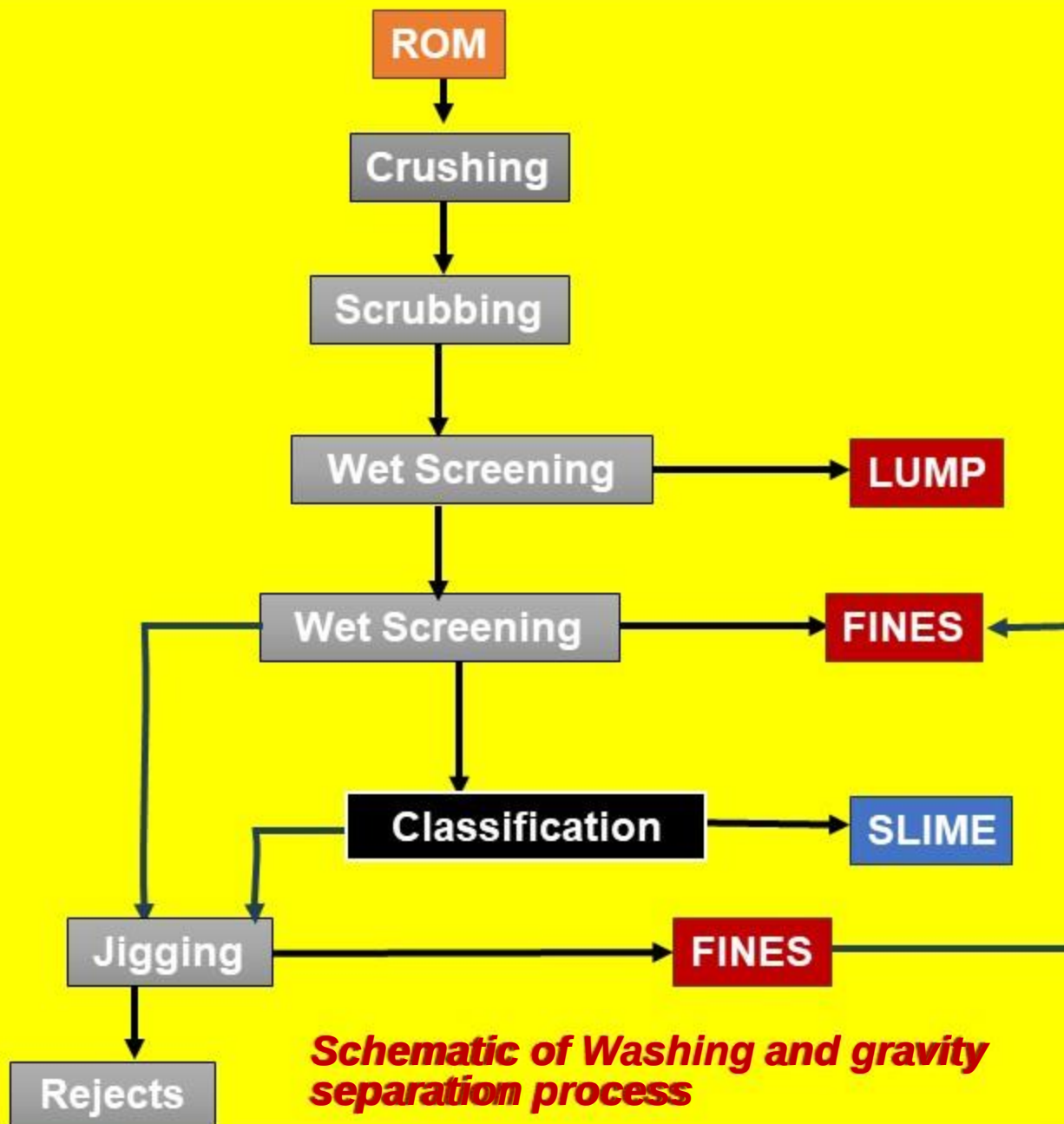
i) **Dry screening process** (for high quality flaky ore and blue dust)

ii) **Wet screen classification** (for ore types having gangue materials)

- ❑ Iron processing depend mainly on the type of ROM ore feed and optimum product. **Dry screening** into **Lumps** and **Fines** is practiced for high quality flaky ore and blue dust, because, **if wet treatment** is used, a substantial part of good quality material is rejected in the form of **Slimes**. Another advantage of dry screening is dry screened fines also retain ultra-fines particle may later be used in **Sintering**.
- ❑ Ore types having gangue material which strictly adhere to the useful metal surface are subjected to wet screening → classification or scrubbing → wet screening → classification.
- ❑ Mineral processing plants at mines like Aswan/Egypt and many mines like Barsua, Bolani, Bailadila, Donamalai, Dalli, Gua, Kiriburu, Meghahatuburu, Noamundi and Rajhara –India **use dry screening for direct ore mined from the face**.
- ❑ **Scrubbers** are being used in Barsua, Bolani, Dalli and Noamundi **for better recovery**.
- ❑ Noamundi mine have **hydro cyclones** , **screw classifiers** for **better beneficiation process**.
- ❑ Various techniques and methods generally being used in the iron ore processing are schematically shown in the figures below↓.



Schematic shows various Beneficiation techniques and methods generally being used in the iron ore processing.



Schematic of Washing and gravity separation process

Lump Iron Ores

- ❑ Lump ore is one of the key ferrous materials used in the blast furnace ironmaking process, typically accounting for 5-25 percent of the blast furnace ferrous burden.
- ❑ Like other ferrous materials, the characteristics of lump ore have an important impact on stable operation and productivity of a blast furnace (BF). As the cold burden lands on the top of the hot burden in the furnace and descends, it is inevitably subjected to thermal shock, which will likely lead to breakdown of the lump particles and affect the gas permeability inside the blast furnace (BF) adversely (Lu, et al., 2017).
- ❑ **Log washers** are being used to produce better and stable quality lumps of iron ore in Sesa Goa mine- India.



- ❑ Slime is an ultra-fine materials (<50 microns) of low grade iron ore, having >2% Al_2O_3 . Two distinct mineral constituent of Alumina in Egyptian and Indian iron ore slimes are gibbsite (hydrated aluminium oxides) and kaolinite. Exact amount of alumina has not been quantified till date according to liberation studies a significant portion of alumina is present in the liberated form and so it is possible to separate them using physical methods. These slimes are dumped which causes environmental hazards.
- ❑ If pellets are produced directly from such fines/slime ore, they become high in alumina which is undesirable in Blast Furnace (BF) as high alumina content will adversely affect the pellet properties, typically measured by Reduction Degradation Index and Reducibility Index (RI). **A drop in alumina content can improve these properties and also reduce the coke consumption in Blast Furnace (BF).**
- ❑ Tailings contain harmful material like iron sulphide which is primary source for **Acid Mine Drainage (AMD)**.
- ❑ **Slag viscosity in blast furnace is increased due to high alumina content. This leads to increase in metal loss in slag, increase in thermal requirement and thus makes the Blast furnace operation more tough.**

What is the difference between fines and lump iron ore?

- ❑ When ore is crushed and split into lump and fines components, there are systematic differences between the lump and fines compositions. The lump component is usually (but not always) richer in iron, and with less silica and alumina, than the fines product.

5.3. Beneficiation and Processing of Iron Ore...(Cont.)

vi) Pelletizing

- ❖ Pelletizing is another method of agglomerating iron ore fines into larger, more uniform pellets. The fine particles are rolled into small balls or pellets, which are then hardened through heat treatment.
- ❖ Pelletizing enhances the physical and metallurgical properties of iron ore, making it suitable for use in steelmaking.

5.4. Mine Tailings

- ❑ For every **one tonne** of iron ore concentrate produced approximately **2.5–3.0 tonnes** of iron ore tailings will be discharged.
- ❑ Statistics show that there are 130 million tonnes of iron ore discharged every year. If, for example, the mine tailings contain an average of approximately 11% iron there would be approximately 1.41 million tonnes of iron wasted annually.
 - These tailings are also high in other useful metals such as copper, nickel, and cobalt, and they can be used for road-building materials like pavement and filler and building materials such as cement, low-grade glass, and wall materials.
 - While tailings are a relatively low-grade ore, they are also inexpensive to collect as they do not have to be mined. **Because of this companies (such as Magnetation) have started reclamation projects** where *they use iron ore tailings as a source of metallic iron.*
- ❑ The two main methods of recycling iron from iron ore tailings are ***Magnetizing Roasting*** and ***Direct Reduction***.
 - i) ***Magnetizing Roasting*** uses temperatures between 700 and 900°C for a time of under 1 hour to produce an iron concentrate (Fe_3O_4) to be used for iron smelting. For magnetizing roasting it is important to have a reducing atmosphere to prevent oxidization and the formation of Fe_2O_3 because it is harder to separate as it is less magnetic
 - ii) ***Direct Reduction*** uses hotter temperatures of over 1000°C and longer times of 2–5 hours. Direct reduction is used to produce sponge iron (Fe) to be used for steel making. Direct reduction requires more energy as the temperatures are higher and the time is longer and it requires more reducing agent than magnetizing roasting.

5.5. Exporting Iron Ore

- ❑ Once the iron ore has been processed into the desired form (such as concentrate, sinter, or pellets), it is ready for export to steel-producing countries around the world.
- ❑ Iron ore is transported via **rail, trucks, or ships** to reach its destination, where it will be used in steel mills for further processing.

6. ENVIRONMENTAL IMPACT OF IRON ORE MINING

- ❑ Iron ore mining, like any other large-scale mining activity, can have significant environmental impacts. It is important to manage and mitigate these impacts to minimize harm to the environment and surrounding communities.
- ❑ Some of the key environmental concerns associated with iron ore mining include *water pollution, air pollution, deforestation, and soil erosion*.

i) Water Pollution

- Water pollution can occur during various stages of iron ore mining, including the excavation, processing, and disposal of waste materials. Chemicals used in processing, such as cyanide or sulfuric acid, can contaminate nearby water sources if not properly managed.
- Additionally, the accumulation of sediment and runoff from mining operations can lead to increased turbidity and degradation of water quality.

ii) Air Pollution

- Air pollution can result from the emissions of dust and particulate matter during mining, crushing, and transportation activities. These airborne particles can have detrimental effects on both human health and the environment.
- Iron ore mining operations must employ dust suppression measures and adopt technologies that reduce emissions to mitigate air pollution.

iii) Deforestation

- The establishment of mining infrastructure, including roads, railways, and processing plants, can lead to deforestation and habitat destruction. Clearing large areas of land for mining purposes can disrupt ecosystems, displace wildlife, and contribute to the loss of biodiversity.
- Proper land management and reclamation strategies are crucial to minimize deforestation and restore affected areas.

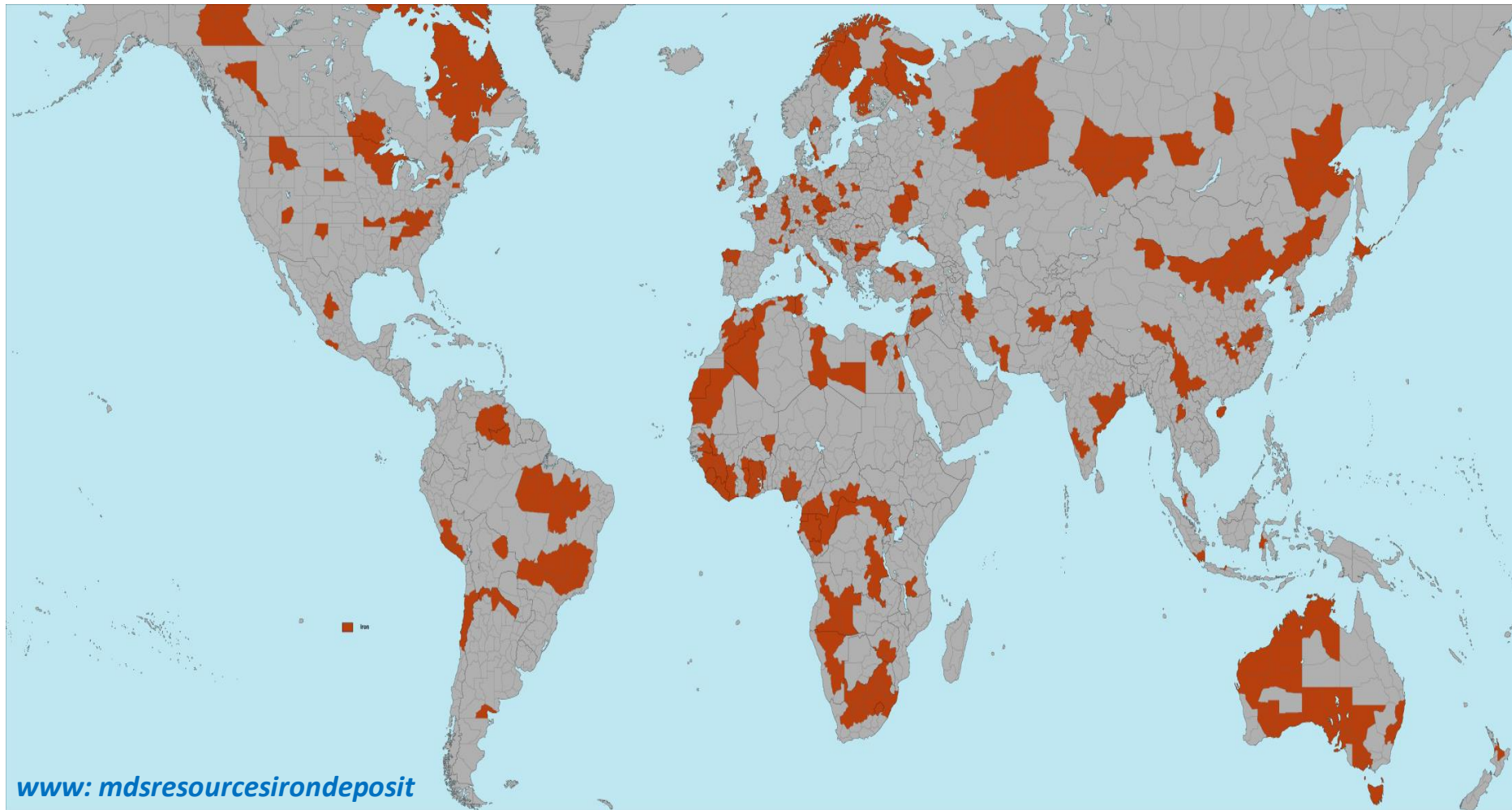
iv) Soil Erosion

- Mining activities can cause soil erosion, especially in open-pit mining, where large areas of land are excavated. The removal of vegetation and topsoil exposes the underlying soil to erosion by wind and water.
- Soil erosion can lead to the loss of fertile land, degradation of ecosystems, and sedimentation in nearby water bodies. Implementing erosion control measures, such as re-vegetation and terracing, can help prevent soil erosion and promote sustainable mining practices.

Mitigating Environmental Impact

- To address the environmental impact of iron ore mining, industry stakeholders, governments, and communities are increasingly focused on implementing sustainable mining practices.
- These practices aim to minimize the environmental footprint of mining operations by adopting technologies that reduce water and energy consumption, improving waste management and reclamation strategies, and enhancing biodiversity conservation efforts.

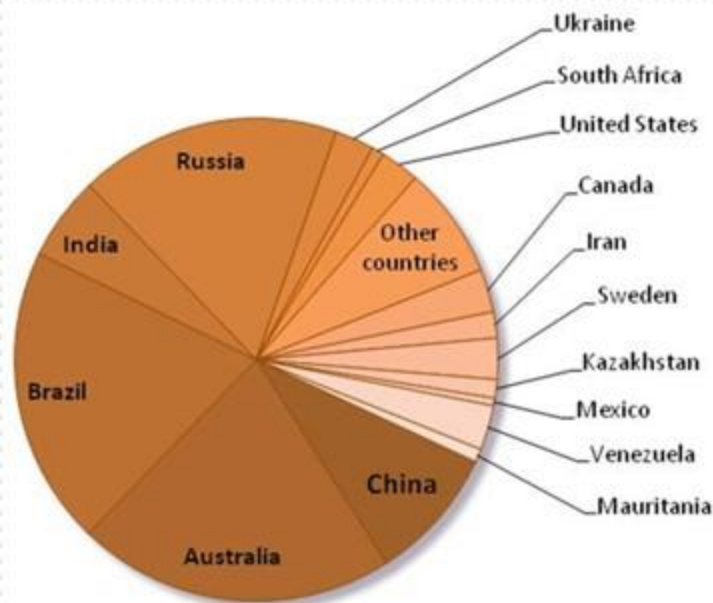
7. WORLD RESOURCES IRON DEPOSITS



- ☐ Iron ore is mined in about 50 countries.
- ☐ The seven largest of these producing countries account for about three-quarters of total world production.
- ☐ Among the largest iron ore producing nations are **Australia, Brazil, China, Russia, India and the USA.**

7.1. Iron Ore Reserves

- Iron ore deposits are found in various parts of the world, with significant reserves located in countries such as **Australia, Brazil, China, India, Russia, and South Africa, among others.**
- These deposits may occur in the form of *hematite, magnetite, limonite, or siderite ores*, each with different iron content and properties.



World reserves of crude iron ore, by country, 2021 (p)

Ranking	Country	Million tonnes	Percentage of total
1	Australia	51,000	28.5%
2	Brazil	34,000	19.0%
3	Russia	25,000	13.9%
4	China	20,000	11.2%
5	Ukraine	6,500	3.6%
6	Canada	6,000	3.3%
7	India	5,500	3.1%
8	United States	3,000	1.7%
9	Iran	2,700	1.5%
10	Peru	2,600	1.5%
-	Other countries	22,930	12.8%
Total		179,230	100.0%

7.2. World Mine Production of Iron Ore

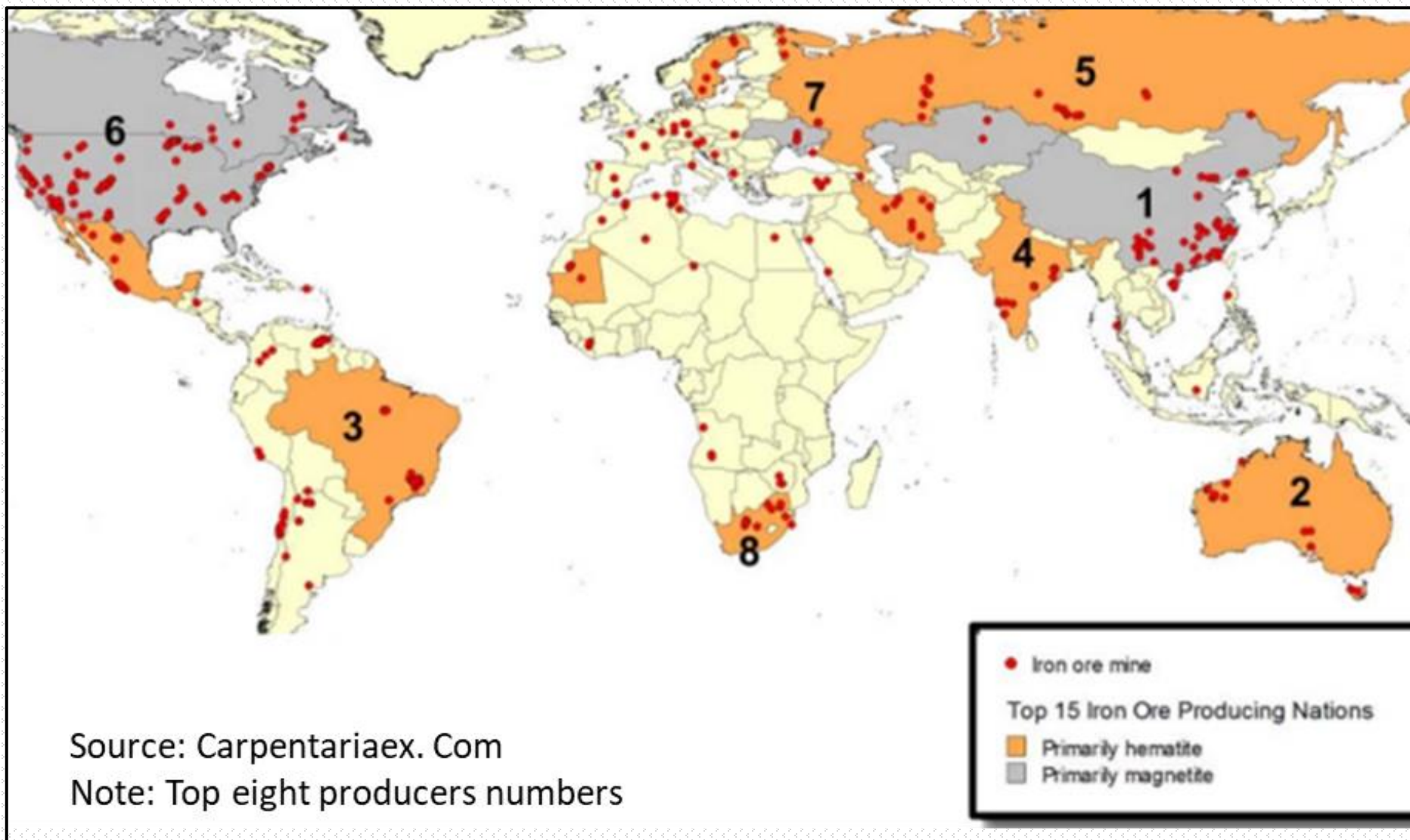
International context

- ❑ Global production of iron ore is estimated to have been 2,537 million tonnes in 2021.
- ❑ Australia is the leading global producer of iron ore, accounting for 34.6% of the total production.
- ❑ The top three iron ore-producing countries accounted for 63% of global production.

World mine production of iron ore, by country, 2021 (p)

Ranking	Country	Million tonnes	Percentage of total
1	Australia	900	34.6%
2	Brazil	380	14.6%
3	China	360	13.8%
4	India	240	9.2%
5	Russia	100	3.8%
6	Ukraine	81	3.1%
7	Kazakhstan	64	2.5%
8	South Africa	61	2.4%
9	Canada	57	2.3%
10	Iran	50	1.9%
-	Other countries	244	9.6%
Total		2,537	100.0%

Top 15 Iron Producing Countries



7.2. World Mine Production of Iron Ore

- ❑ Iron is the world's most commonly used metal-steel, of which iron ore is the key ingredient, representing almost 95% of all metal used per year. It is used primarily in structures, ships, automobiles, and machinery.
- ❑ Iron-rich rocks are common worldwide, but ore-grade commercial mining-operations are dominated by the countries listed in the next table.
 - The major constraint to economics for iron ore deposits is not necessarily the grade or size of the deposits, because it is not particularly hard to geologically prove enough tonnage of the rocks exist.
 - *The main constraint is the position of the iron ore relative to market, the cost of rail infrastructure to get it to market and the energy cost required to do so.*
- ❑ Mining iron ore is a high-volume, low-margin business, as the value of iron is significantly lower than base metals.
 - It is highly capital intensive, and requires significant investment in infrastructure such as rail in order to transport the ore from the mine to a freight ship. For these reasons, iron ore production is concentrated in the hands of a few major players.

7.2. World Mine Production of Iron Ore.....(Cont.)

- ❑ World production averages **two billion metric tonnes of raw ore annually**. The world's largest producer of iron ore is the **Brazilian mining corporation Vale**, followed by **Australian companies Rio Tinto Group and BHP**. A further Australian supplier, Fortescue Metals Group Ltd, has helped bring Australia's production to first in the world.
- ❑ The seaborne trade in iron ore-that is, iron ore to be shipped to other countries-was 849 million tonnes in 2004.
 - Australia and Brazil dominate the seaborne trade, with 72% of the market.
 - BHP, Rio and Vale control 66% of this market between them.
- ❑ In Australia, iron ore is won from three main sources: ***Pisolite "Channel Iron Deposit"*** ore derived by mechanical erosion of primary banded-iron formations and accumulated in alluvial channels such as at Pannawonica, Western Australia; and the dominant metasomatically-altered banded iron formation-related ores such as at Newman, the Chichester Range, the Hamersley Range and Koolyanobbing, Western Australia. Other types of ore are coming to the fore recently, such as oxidized ferruginous hardcaps, for instance laterite iron ore deposits near Lake Argyle in Western Australia.
- ❑ The total recoverable reserves of iron ore in India are about 9,602 million tonnes of hematite and 3,408 million tonnes of magnetite. Chhattisgarh, Madhya Pradesh, Karnataka, Jharkhand, Odisha, Goa, Maharashtra, Andhra Pradesh, Kerala, Rajasthan and Tamil Nadu are the principal Indian producers of iron ore.

7.3. The 20 Largest Iron Ore Mines in the World (2021)

Property	Location	Majority Ownership	Number of Operating Mines	2021 Annualized Production (Mt)	2020 Production (Mt)
Hamersley Mines and Channar	Australia	Rio Tinto	11	207.04	219.86
Sistema Norte	Brazil	Vale	5	185.08	192.27
Fortescue Operations	Australia	Fortescue Metals	4	182.93	180.30
Area C and Yandi	Australia	BHP Group	2	147.81	141.49
Newman and Jimblebar	Australia	BHP Group	2	132.36	146.82
Hope Downs and West Angelas	Australia	Rio Tinto	2	81.15	83.25
Vargem Grande	Brazil	Vale	5	30.93	25.10
Sishen	South Africa	Kumba Iron Ore	1	28.64	25.35
Itabira	Brazil	Vale	3	27.61	23.91
LKAB operations	Sweden	LKAB	3	27.11	27.10
Robe Valley	Australia	Rio Tinto	2	25.96	30.29
Paraopeba	Brazil	Vale	4	24.11	23.30
Minas Rio	Brazil	Anglo American	1	23.47	24.08
Mont-Wright	Canada	ArcelorMittal	1	23.20	23.20
Lebedinsky	Russia	Metalloinvest	1	22.31	22.06
Mariana	Brazil	Vale	4	21.03	17.72
Stoilensky	Russia	Novolipetsk Steel	1	19.80	18.51
Minas Centrais	Brazil	Vale	1	19.52	15.65
Mikhailovsky	Russia	Metalloinvest	1	18.63	18.37
Carol Lake	Canada	Rio Tinto	1	16.41	17.77

*Annualized figures based on year to end-of-Q3 company production reports

**No 2021 data is available for LKAB, Mont-Wright

Source: Miningintelligence

7.3. The 20 Largest Iron Ore Mines in The World (2021)

1. Hammersly Mines and Channar

- In first place is Rio Tinto's Hamersley and Channar Mines complex with 11 operating mines in the Pilbara region of Australia. Based on figures provided through the end of Q-3 2021, 207.04 million tonnes (mt) of iron ore are forecast for the year. The world's second largest miner came under fire when an inquiry found that Hamersley Iron failed to protect artifacts belonging to the Wintawari Guruma Aboriginal Corporation after last year's explosion. The fallout would be the resignation of top executives, including the CEO, and replacement by leaders with a stronger focus on sustainability..

2. Northern System

- In second place is Vale's Northern System in the state of Pará, with 5 mines in operation with a projected production of 185.08mt of ore in the year 2021. In the second quarter, Vale announced and the restart of operations at Serra Leste in the North System. The expansion includes the refurbishment of an iron ore plant that is scheduled to increase Serra Leste's processing capacity to 10 million tons per year by early 2023.

3. Fortescue Operations

- In third place are Fortescue Metals' operations in Australia's iron ore-rich Pilbara region, which includes the Chichester, Solomon, Hedland and Western mining centers, where 182.93mt of iron ore is expected in 2021. In December, the world's fourth largest mining company announced that chief executive Elizabeth Gaines will step down as the company extends its transition from an iron ore producer to an energy and green resources company.

4. Area C and Yandi

- The world's largest mining company, BHP, comes in fourth place with its Area C and Yandi mines in Australia. In 2021, 147.81mt of iron ore was expected. BHP has the green light to go ahead with a planned expansion of its operations in the Pilbara region, consolidating its presence over the next 50 to 100 years.

5. Newman and Jimblebar

- BHP's Newman and Jimblebar mines in Pilbara, Australia, rank 5th with a total of 132.36mt of iron ore projected for the year 2021. A few years ago, a \$3.6 billion expansion at Newman and Jimblebar significantly increased the capacity of BHP Pilbara's supply chain.



Rio Tinto's iron ore operations in the Pilbara began in 1966, with the Mount Tom Price mine (pictured) part of Hamersley mine complex, opening that year. (Image courtesy of [Wikipedia](https://en.wikipedia.org/wiki/Mount_Tom_Price_mine).)

7.3. The 20 Largest Iron Ore Mines in The World (2021)

6. Hope Downs and West Angelas

- Rio Tinto's Hope Downs and West Angelas mines in Pilbara, Australia, would have produced 81.15 million tons of iron ore by 2021. Rio Tinto expanded its fleet of operatorless trucks, introducing 15 autonomous vehicles in West Angeles in 2018.

7. Vargem Grande

- In seventh place, Vale's Vargem Grande complex with five mines located in Brazil was projected to produce 30.93mt by the year 2021. In a move to reduce dependence on dams, Vale has begun operations at a tailing's filtration plant at the Vargem Grande complex, the first of four that the company plans to install at sites in Minas Gerais at a cost of \$2.3 billion between 2020 and 2024.

8. Sishen

- Kumba Iron Ore's Sishen complex in South Africa's Northern Cape province is ranked eighth. Based on annualized figures, it was expected to produce 28.64mt of iron ore in 2021. Kumba said in February that it planned to spend 3.6 billion rand to extend the life of the Sishen mine to 2039 through the ultra-dense media separation project, scheduled to start in the second half of 2023.

9. Itabira

- Back in Brazil, Vale's three operating mines in Itabira were forecast to produce 27.61mt of iron ore by 2021. Itabira is the site of Vale's first mine. It is also one of the ten cities affected by the collapse of the tailings dam in Brumadinho.

10. LKAB Operations

- LKAB Operations in Sweden, owned by LKAB, has three operating mines slated to produce 27.11mt of iron ore by 2021. LKAB upgraded its ore processing capacity a few years ago, adding an RMD system as part of an ongoing program of expansion in an area that has been home to iron ore mining since the mid-18th century.

11. Robe Valley

- Rio Tinto's Robe Valley operates at Pilbara in Australia, the Mesa A and Warramboos mines were projected to produce 25.96mt of iron ore in 2021.

12. Paraopeba

- Vale's operations in Paraopeba, Minas Gerais, have four operating mines slated for 24.11mt of ore by 2021. In 2019, it began cleaning up the Paraopeba River, which was contaminated that year after a dam broke at the Feijão mine.

13. Minas Rio

- Anglo American's giant Minas Rio mine in Brazil was scheduled to produce 23.47 mt of iron ore in 2021. Two years ago, Anglo brought its giant Minas Rio mine up to full capacity after the country's authorities granted it a key permit for the use of an expanded tailings dam at the site.





Kumba Iron Ore is South Africa's top producer of the steel-making ingredient.
(Image courtesy of [Kumba Iron Ore](#))

7.3. The 20 Largest Iron Ore Mines in The World (2021)

14. Mont Wright

- ArcelorMittal's Mont Wright mine in Quebec, Canada, produced 23.2mt of iron ore in 2021. ArcelorMittal, the world's largest steelmaker, reported its highest quarterly earnings in 13 years in July, beating expectations, and announcing a new share buyback.

15. Lebedinsky

- In Russia, Metalloinvest's Lebedinsky mine was scheduled to produce 22.3mt in 2021. Metalloinvest claims the world's largest iron ore company by JORC reserves, as its economically recoverable iron ore reserves total 15.4 billion tons.

16. Mariana

- Vale's Mariana mining hub in Brazil consists of four mines, and is expected to produce 21.03mt of iron ore in 2021.

17. Stoilensky

- Novolipetsk Steel Company's Stoilensky mine in Russia is number 16 in terms of production, forecast to produce 19.80mt of iron ore in 2021.

18. Minas Centrais

- Vale's Minas Centrais mine in Brazil is expected to produce 19.52mt of iron ore in 2021.

19. Mikhailovsky GOK

- Metalloinvest's Mikhailovsky GOK is expected to produce 18.63mt of ore in 2021. Mikhailovsky GOK was the first in Russia to use hematite concentrate, a technology based on the use of flotation plants that reduces damming tailings and increases production without additional iron ore extraction.

20. Carol Lake

- Completing the list is Rio Tinto's Carol Lake, Canada's largest open-pit iron ore mine, located in the eastern province of Newfoundland (Newfoundland and Labrador). Carol Lake was projected to produce 16.41mt of iron ore in 2021.

7.4. Largest Iron Ore Producers

An overwhelming proportion of the world's iron ore mining happens in Australia. As shown by the graph below, Australia generated an estimated 880Mt in 2023, with Brazil coming second at a significantly lower 410Mt.

Vale

- ❖ In terms of leading iron ore producing companies in 2021, Brazilian mining giant Vale (BOVESPA: VALE3) salvaged the top spot again by producing an estimated 319.6M tonnes that year. One of Vale's most significant iron ore mines is the Carajás Mine located in Para in Northern Brazil, holding 7.27B tonnes of iron ore reserves (proven and probable).

Rio Tinto

- ❖ The second largest iron ore producer came out as Rio Tinto (ASX: RIO), generating an estimated 276.6M tonnes in 2021. Rio Tinto has an iron ore mining network located in the Pilbara region of Western Australia, comprising 16 substantially sized mines.

BHP (formerly BHP Billiton)

- ❖ Similar to Rio Tinto iron ore, BHP (ASX: BHP) currently owns five mines in Western Australia's Pilbara region. The company produced an estimated 245.4M tonnes of Iron ore in 2021. BHP is also responsible for implanting Australia's largest new iron ore mine in more than 50 years, known as the South Flank mine.

Fortescue Metals

- ❖ Finally, in fourth place is Australian-based miner Fortescue Metals. The Fortescue Metals Group (ASX: FMG) produced 84.7M tonnes of iron ore in 2021. Fortescue Metals Group's largest Christmas Creek Mine is also in Western Australia.

Desirable Pilbara blend in the Mid-West

- ❖ The Pilbara region is Australia's mining powerhouse and contributes considerably to the country's national wealth. Its iron ore and liquefied natural gas industries are valued at over AUS\$70B, representing more than 70% of mineral and energy production in Western Australia. Many miners are drawn to Pilbara's rusty red cliff's natural resources.
- ❖ Australian producers of iron ore experience a logistical advantage over their Brazilian rivals. This is due to advanced port facilities and only 12 days to complete iron ore exports to China compared with around 45 days from Brazil.

7.5. Platts Iron Ore Index (or IODEX (IODBZ00))

- ❑ Platts Iron Ore Index, or IODEX (IODBZ00), is a benchmark assessment by S&P Global Commodity Insights of the spot price of physical iron ore.
- ❑ The assessment is based on a standard specification of iron ore fines with 62% iron, 2.25% alumina, 4% silica and 0.09% phosphorus, among other gangue elements.

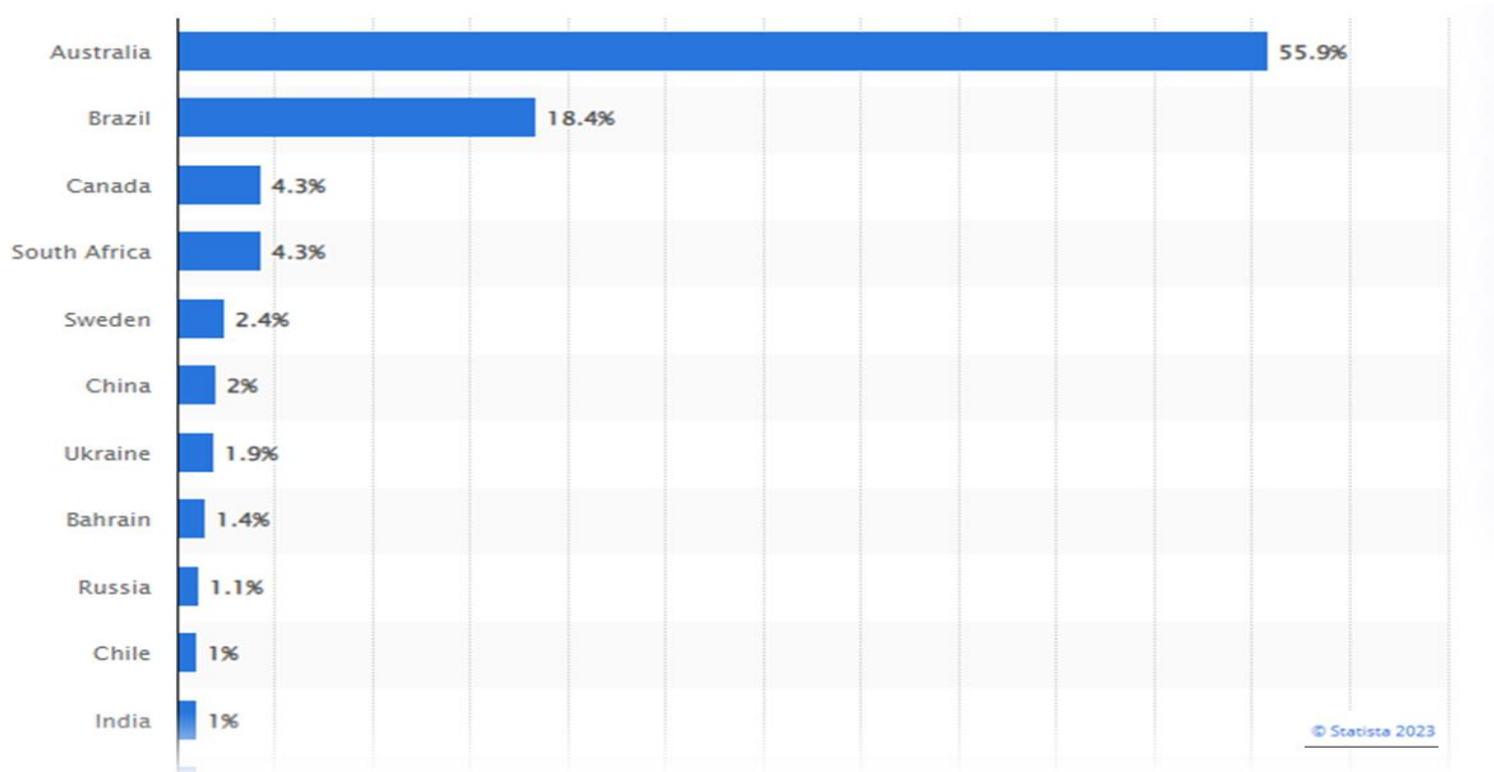
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8. WORLD CONSUMPTION OF IRON ORE

- ❑ World consumption of iron ore grows 10% per annum on average with the main consumers being *China, Japan, Korea, the United States and the European Union*.
- ❑ **China** is currently the largest consumer of iron ore, which translates to be the world's largest steel producing country. It is also the largest importer, buying 52% of the seaborne trade in iron ore in 2004.
- ❑ China is followed by Japan and Korea, which consume a significant amount of raw iron ore and metallurgical coal.
- ❑ In 2006, China produced 360 million tonnes of iron ore, with an annual growth of 38%.

9. DISTRIBUTION OF GLOBAL IRON ORE EXPORTS in 2022, by major country

- ❑ Australia was by far the country exporting the largest amount of iron ore worldwide in 2022.
 - Some 56% of global iron ore exports came from Australia, worth 87.7 billion U.S. dollars that year.
 - Australia is able to export the most iron ore worldwide due to the fact that it also has the world's largest reserves of iron ore.



10. ABOUT IRON ORE MARKET

- ❑ The biggest *Iron Ore Mines* are located in *Brazil, Australia, Republic of Congo, Guinea, and South Africa*
- ❑ **China** is the leading manufacturer of Iron Ore in global market supplying about a *Half of World* total; together with Australia and Brazil they account for 2/3 of global output
 - with the development of new cost-efficient mines worldwide China had to close a number of local low-grade iron mines
 - future of global iron ore industry is dependent on the situation in the world steel industry and greatly influenced by China.
 - global iron ore production is projected to grow at 2-2.3% annually as a result of substitution of China's domestic production of iron ore with imports, particularly from Australia and Brazil.

11. CONCLUSION

- ❑ Iron ore mining plays a critical role in supplying the raw material necessary for steel production, supporting various industries and economic development worldwide.
- ❑ From the extraction of iron ore to its processing and eventual export, each stage of the mining process requires careful planning, technological advancements, and environmental considerations.
- ❑ By adopting sustainable mining practices and mitigating environmental impacts, the future of iron ore mining can be aligned with the principles of responsible resource utilization and environmental stewardship.

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